Soyo discharge controller (bat2 -> grid)

The Soyo is a grid-tie micro-inverter that discharges the EBox battery (bat2) into the house to keep grid import near zero — the counterpart to the EBox charger (fox2db). Setpoint w in [0,900] W, cycle 60 s, sent over RS485 every 3 s.

Classification

Gated proportional controller with night feed-forward and saturation. P-term on grid import (target pcc=0), no I/D term. Five gates force w=0 (safety / no simultaneous charge+discharge).

Inputs and output

```
Inputs :  pcc   grid exchange [W]  (< 0 = import)
          st    EBox state 0..7    (charger state)
          soc2  bat2 SOC [%]
          ebox  measured EBox load [W]
          h     hour (day/night)
Output :  w     Soyo setpoint 0..900 W  (RS485 frame)
```

Constants

```
k_p (P gain) ............. 1.01
B_n (night base load) .... 468 W   (feed-forward)
B_d (day idle) ........... 10 W
w_max (saturation) ....... 900 W
import threshold ......... -100 W
surplus gate ............. 200 W
deep-discharge guard ..... soc2 < 9 %
night .................... h < 6 or h >= 20
TX watchdog .............. setpoint > 90 s old -> 0
```

Mathematical model

Gate cascade — w = 0 if any condition holds

```
G1  stale        (inverter data > 3 min old)
G2  st != 0      (EBox charging -> no simultaneous discharge)
G3  0 <= soc2 < 9   (deep-discharge guard)
G4  ebox > 200   (bat2 charging)
G5  pcc  > 200   (PV surplus -> spare bat2)
```

Active law (gate open)

$$w = \mathrm{sat}_{[0,\,900]}\big(k_p\,(-\mathrm{pcc}) + B_n\,\mathbf{1}[\mathrm{night}]\big) \quad \mathrm{pcc} < -100$$

$$w = B_n\,\mathbf{1}[\mathrm{night}] + B_d\,\mathbf{1}[\mathrm{day}] \quad -100 \le \mathrm{pcc} \le 200$$

The P-term with k_p~1 is a unity feed-forward cancellation: inject as much as is currently being imported. B_n=468 W is the night base load as feed-forward. The dead-band [-100, 200] W prevents hunting around zero.

Constants

$$k_p = 1.01,\ B_n = 468,\ B_d = 10,\ w_{\mathrm{max}} = 900\ [\mathrm{W}]$$

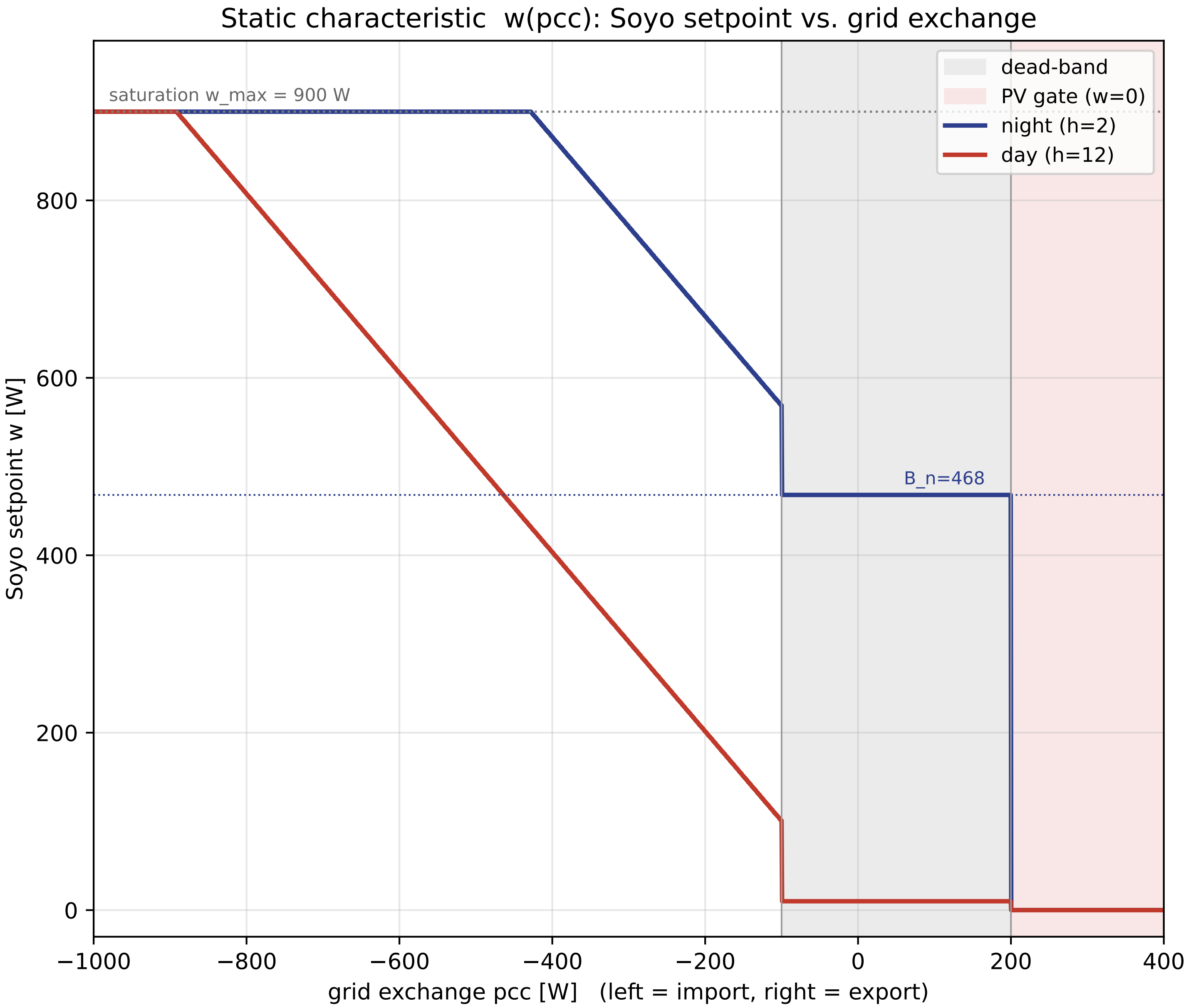
TX watchdog (RS485, every 3 s)

$$w_{tx} = w \cdot \mathbf{1}[\Delta t \le 90\,\mathrm{s}]$$

If the setpoint is not refreshed in time (controller failure), the RS485 side sends 0 W -> the Soyo stops discharging (fail-safe).

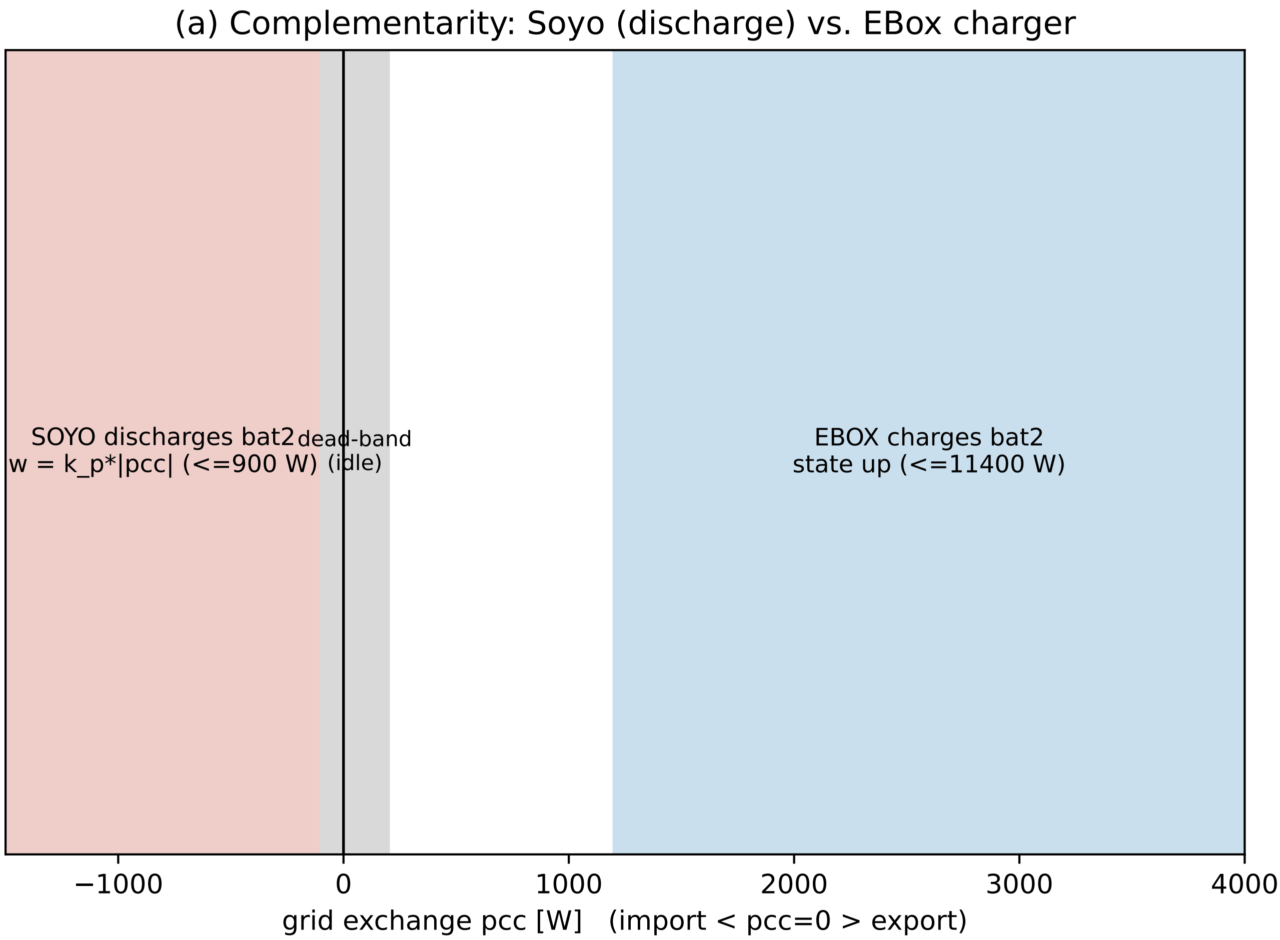
Protocol frame (8 bytes)

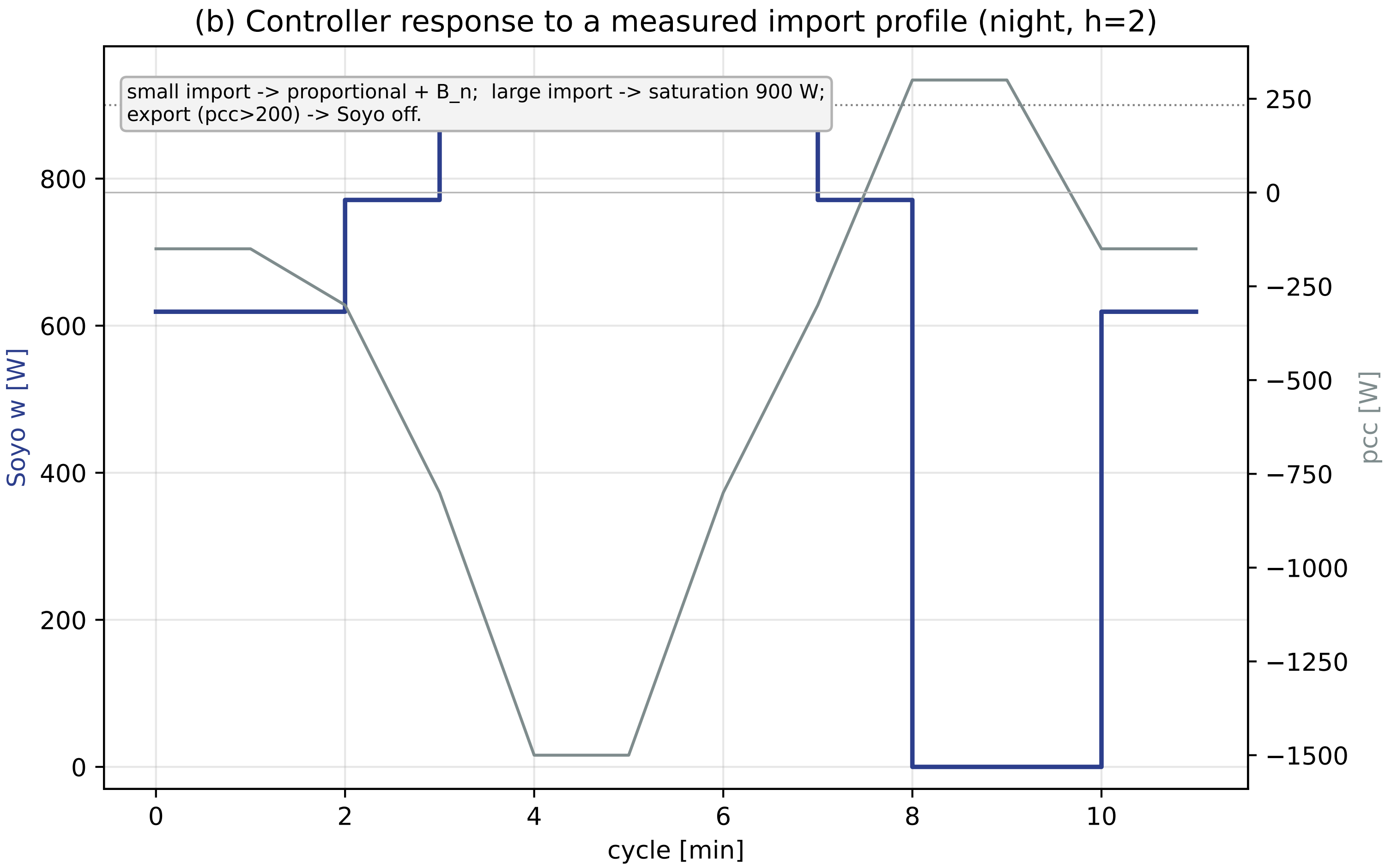
```
24 56 00 21  ph pl  80  crc      ph=(w>>8)&0xFF, pl=w&0xFF
crc = (264 - ph - pl) & 0xFF
```




```
Regions (left to right):
  - pcc < -100 W : proportional, w = k_p*|pcc| (+B_n at night), up to saturation 900 W
  - -100..200 W  : dead-band - only base load (468 W night) resp. 10 W day
  - pcc > 200 W  : PV surplus -> Soyo off (w=0), spare bat2

The curve is the complete (memoryless) definition of the P-controller; the gates
G1-G4 (stale / EBox on / SOC<9% / bat2 charging) additionally force it to 0.
```


(a) Complementarity: Soyo (discharge) vs. EBox charger
SOYO discharges bat2
w = k_p*|pcc| (<=900 W)
dead-band
(idle)
EBOX charges bat2
state up (<=11400 W)
−1000
0
1000
2000
3000
4000
grid exchange pcc [W]   (import < pcc=0 > export)


(b) Controller response to a measured import profile (night, h=2)
small import -> proportional + B_n; large import -> saturation 900 W;
export (pcc>200) -> Soyo off.
Soyo w [W]
0
200
400
600
800
pcc [W]
250
0
−250
−500
−750
−1000
−1250
−1500
0
2
4
6
8
10
cycle [min]

Classification & artifacts

Control-theoretic character

Pure P-controller on the import error e=-pcc (target pcc=0) with gain k_p~1 (quasi dead-beat per cycle), feed-forward night base load, saturation and dead-band. No integrator -> no windup; the controller relies on re-measuring pcc every 60 s cycle.

Interplay with the EBox charger

Both controllers tile the pcc axis around 0 and are mutually exclusive: the gates st!=0, ebox>200 and pcc>200 prevent bat2 from charging and discharging at the same time. Soyo covers small/medium loads (base load, <=900 W); the EBox charger absorbs large PV surplus.

Limits

Due to saturation at 900 W the Soyo covers only base load; larger consumers are carried by the Sofar battery (bat1) and the grid. The night feed-forward of 468 W damps hunting at the dead-band edge at night.

Artifacts

```
sofar_waveshare.yaml   Soyo calculation (ESPHome, interval 60s + RS485)
soyo_model.py          math reference + self-tests
make_soyo_pdf*.py      this document
```